When Schema Lexicons Fail: Diagnosing and Repairing Entity-Layer Collapse in a Production Knowledge Base

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ABSTRACT

In dictionary-driven entity enrichment, the schema lexicon effectively serves as the extraction model. We report a production incident in which a category-word lexicon silently collapsed the entity layer of a 61,137-document scholarly knowledge base: 575 type-level entities ("method", "energy") carrying ∼19.2M links while every job-level monitor stayed green. We diagnose the failure across three layers (extraction mechanism, lexicon provenance, monitoring blindness), repair it with a three-source corpus-presence-attested lexicon (3,702 terms), rebuild the entity layer under production load, and derive scale-invariant quality monitors. A controlled experiment that fixes code and corpus and varies *only the lexicon* reproduces the collapse signature alone (109±5 entities, top-10 concentration 0.513±0.016, 1.85±0.05 concepts/document) and isolates the repair effect (692±27, 0.300±0.008, 3.89±0.07; mean±sd over ten random resamples of the same corpus); the rebuilt layer carries 1,871 entities, 25.1M links, and a functioning relation layer; distinct concepts per document rise 15.7→37.8 with document coverage 48.5%→98.5%; a risk-stratified 300-link audit (187 distinct entities; a stratum-mixed rate, not a link-uniform estimate) reads 0.84 overall — 0.95 on domain terms, 0.58 on short ambiguous symbols. The generic tail survives repair (9% of links sit on the single ambiguous symbol "be"), the isolated-experiment metrics are structural rather than semantic, retrieval-facing gains are not attributable to the lexicon alone (removing the direct entity channels leaves @10 metrics unchanged), audit is model-assisted rather than human-adjudicated, and the concentration alarm bands are candidate thresholds read post-hoc from 13 libraries — limitations stated up front.

KEYWORDS

knowledge base construction schema quality terminology extraction entity linking production incident observability

1 INTRODUCTION

Scholarly knowledge platforms increasingly generate subject schemas with LLM assistance: a batch tool drafts entity types with a handful of aliases each, a validator checks structural well-formedness, and the schema is bound to knowledge bases as configuration [5]. When the enrichment pipeline then extracts mentions by *dictionary scan*, the schema's alias list becomes the extraction surface — the lexicon is not descriptive metadata but operative infrastructure.

The failure we report: a 61,137-document nuclear-electronics knowledge base accumulated ∼19.2M entity links onto **575 canonical entities**, every one a bare type word ("method", "parameter", "energy"); a 13,408-document physics library showed the same failure pattern at 2,018 entities. Every operational signal stayed green — jobs done, throughput nominal, link volume reading as growth. The graph had degenerated into a bipartite type–document index that was row-level indistinguishable from a populated knowledge graph.

We contribute three things. **(1) A failure mode and its causal confirmation**: when extraction is a dictionary scan, a structurally valid schema lexicon acts as the extraction model and can silently collapse the entity layer — established forensically (§3) and then *causally*, by a controlled experiment that fixes code and corpus and varies only the lexicon across ten disjoint corpus resamples (§4). **(2) A corpus-attested lexicon lifecycle** (seed transplant, classified corpus candidates, bounded expansion — corpus presence gated) that repaired both schemas without manual lexicography and survived a full production rebuild (§5–6). **(3) Gold-free entity-quality observability**: concentration-led monitors with candidate thresholds read from 13 production libraries (§7.1), plus a corrected incident log demonstrating that dashboard-derived incident narratives themselves require row-level verification.

2 BACKGROUND AND PRIOR ART

Dictionary- or gazetteer-driven entity recognition is one of the oldest NER traditions [2]; corpus-statistical term recognition goes back to C-value/NC-value [1] — our corpus-candidate source is that program reborn with an LLM classifier. Knowledge-graph refinement [3] and production data validation [4] address adjacent problems. What prior art does not cover is the specific failure locus: when the dictionary effectively *is* the extraction model, a structurally valid but terminologically hollow schema passes every existing gate.

Our pipeline extracts mentions in stage K1 by normalized dictionary scan (CJK substring; ASCII word-boundary with optional plural, case-folded); K2 links mentions to canonical entities by normalized form + type; K3 adjudicates typed relations with an LLM; K4 projects to the graph store. The healthy reference case in the same deployment — hydrogeology, 78 documents, a hand-evolved 3,742-term lexicon, 1,608 entities — establishes plausibility, not causation: domain, scale, and provenance all differ.

3 THE COLLAPSE, FORENSICALLY

Observation (Table 1). During an unrelated storage investigation (2026-10-08), a routine entity-quality sample returned type words where domain concepts were expected.

Mechanism. When the alias set for type *method* contains common words, every occurrence in 7.2M chunks becomes a mention, and K2 folds them onto one entity per (form, type). The mechanism executed faithfully on degenerate input; the dictionaries were the schemas — the batch generator produced a category taxonomy where a terminology was needed [6].

Timeline (compressed): 10-08 discovery → root cause, lexicon tool built/run/committed, teardown + 766,501-event re-enqueue; 10-08 night: silent mis-attribution window (below); 10-09 drain completes, statistics recomputed, two smaller libraries with the

Table 1: The collapse, measured, with snapshot provenance ("bckp." = 2026-10-06 cold backup, mid-enrichment; "disc." = discovery-day).

KB	Entities	Links
nuclear (61,137 docs)	**575** (disc.) / 503 (bckp.)	~**19.2M** (disc.) / 4.81M (bckp.)
physics (13,408 docs)	**2,018**	2.23M (bckp.)
hydro (ref., 78 docs)	1,608	719,515

Table 2: Controlled lexicon-only comparison across ten disjoint corpus resamples (mean±sd).

Arm	Links	Ents	C_{10}	Conc/doc
v2.0 collapse-era	2,969 ± 140	109 ± 5	0.513 ± .016	1.85 ± .05
v2.5 mid	4,823 ± 147	421 ± 25	0.322 ± .009	2.51 ± .05
v2.1 repair	10,042 ± 251	692 ± 27	0.300 ± .008	**3.89 ± .07**
+case-sens. short	8,758 ± 228	665 ± 28	0.265 ± .012	3.54 ± .07
+stop-word block	9,226 ± 251	691 ± 27	**0.252 ± .011**	3.67 ± .07

same failure pattern treated; 10-09/10: retrospective audits (scoped refresh, channel ablation, link audit, fleet-wide concentration) and manuscript. The compression is real; the evaluation is retrospective, not pre-registered.

Why nothing fired. Job-level observability was all green; link volume read as success. The observables that expose the collapse are the concentration of links on the top entities (collapsed: 0.419, measured from the pre-incident backup, versus 0.18 for the reference case) and the name distribution of those entities (pure type vocabulary). Entity cardinality per document — a factor of 2,192 below the reference case (0.0094 vs 20.6) — corroborates, but is gameable in both directions, so it is the secondary signal.

4 IS THE LEXICON THE CAUSE? ISOLATING THE VARIABLE

The repair bundled lexicon, teardown, and fleet changes; this experiment isolates the lexicon alone *before* we present the repair method: a semantic replica of the K1/K2 scan-and-link semantics with *only the dictionary swapped* — v2.0 as committed at library creation (182 scan terms), v2.5 mid-evolution (2,932), v2.1 the repair (3,733; construction in §5), plus two v2.1 governance variants. The replica copies the scanner's matching and folding rules from the same source reading (CJK substring; ASCII word-boundary with optional plural, case-folded; fold key = normalized form + type); it is not instrumented against the live service path, so absolute levels may drift — the experiment's force is the within-frame arm contrast.

Design. Ten seeded, uniformly random resamples of 2,000 chunks each from the library's 5.83M visible chunks — independent draws from a *single* corpus, i.e. a sampling-stability check within one library, not cross-domain evidence — cross-sample disjoint, deduplicated on redraw (an earlier single run had sampled an ordered prefix, which we replace rather than defend). Fixed code, fixed corpus, fixed scanner; the only variable is the dictionary. All metrics are deterministic; Table 2 reports mean±sd across the ten resamples.

Finding 1 — the collapse signature is reproduced by the lexicon alone. Under v2.0, the same code on the same corpus yields 109±5 entities at concentration 0.513±0.016 with 1.85±0.05 concepts per document: the collapse needs no pipeline bug, no data loss, no misconfiguration — only a category-word dictionary.

Finding 2 — the repair effect is isolated and monotone. v2.1 multiplies link-bearing entities ×6.3 and concepts per document ×2.1; generic-head mass falls 0.146→0.058. The production collapsed concentration (C_{10}=0.419) sits between the v2.0 (0.513) and v2.5 (0.322) counterfactuals, consistent with mid-enrichment dictionary evolution. These are *structural* signatures: entity count, concentration, and concepts per document do not by themselves measure terminology recall or sense fit — the only semantic reading in this paper is the link audit of §7.

Finding 3 — the be class is treatable at known cost. Case-sensitive short symbols remove the largest entity at the price of match volume (links −12.8%; C_{10} →0.265); a stop-word block is gentler (links −8.1%, C_{10} 0.252). The deltas measure matching *output*, not true recall: the removed mass mixes errors with correct terms, and the 0.580 pre-intervention judged-valid rate on this class (§7) motivates the treatment without yet demonstrating post-treatment validity — removed-vs-retained link verification is the open evaluation. *Provenance*: decomposing v2.1's link mass by which committed lexicon first contained each term — v2.0-inherited 22.0%, mid-evolution 16.0%, **three-source-new 61.9%** (384 of 686 entities) — is a decomposition, not a per-source causal ablation; independently regenerating single-source lexicons remains future work.

Why 692 entities here vs 1,871 in production (§7)? The isolated frame counts only entities induced by scanning 2,000 chunks (0.034% of the library) under replica folding: rare terms need corpus mass to be attested, and the production layer additionally carries entities from the full corpus and historical merging. The causal claim rests on within-frame comparisons, never on this cross-frame count.

5 CORPUS-ATTESTED LEXICON CONSTRUCTION

The repair lexicon whose causal role §4 isolated is built from three sources, all gated by corpus presence (a term the corpus never attests is not admitted — presence is necessary, not sufficient, for termhood): **seed transplant** (2,284 terms from a healthy near-domain library); **corpus-candidate classification** (top 8,000 unigram/bigram n-grams by chunk frequency over a 3,000-chunk-per-KB sample, frequency ≥5 after boilerplate and stop-word filtering; an LLM classifies each into a schema type or rejects it); **per-type expansion** (≤60 per type). Catch-all types accept no classified noise; cross-type conflicts resolve by specificity. Classifier: Qwen3.6-35B-A3B (AWQ 4-bit, community build; served under a historical alias), temperature 0.2, single idempotent run. Output: nuclear v2.1 = 3,702 terms; physics v1.1 = 1,279.

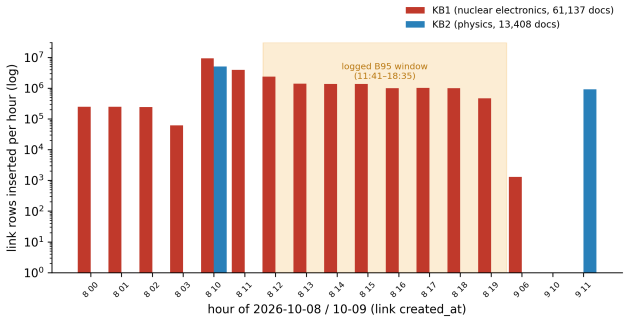


Figure 1: Rebuild telemetry from the link registry's own timestamps. The shaded band is the *logged* silent window; inserts continue through it — the timestamp correction of §6.

6 REBUILD AT SCALE — AND A CORRECTED INCIDENT LOG

Per knowledge base: delete graph vertices with edges, the four relational tables in bounded chunks, and the search-side projection; re-enqueue all documents through four stages. The printed "766,501 jobs" counts *events*, not distinct doc-stages (74,545×4 = 298,180): the surplus is the repair window's re-enqueue waves (silent-empty reset 75,831; zero-effect relations/graph resets ≈149k; watchdog-deletion recovery ≈122k; small libraries 5,772).

Four pipeline pathologies surfaced and were fixed with measured effect — an unbounded projection join that saturated the database (25+ min → seconds once scoped), two link-insertion deadlock classes (deadlocks 1213 → 0), and an $O(N^2)$ alias check the richer lexicon itself amplified (fleet link throughput 8/min → 1,100/min, a window-level aggregate). The archival version carries the full per-pathology accounting.

The fifth pathology, revisited. Our log recorded a parameter-order bug in a single-statement link writer whose rows `INSERT IGNORE` "silently swallowed" for seven hours. A minimal reproduction on the production engine (MySQL 8.4, `STRICT_TRANS_TABLES`, faithful DDL) sharpens the mechanism: the NULL from `UNHEX('alloy')` is **replaced by the implicit default 0x00×16 and the row is inserted** — into a ghost bucket invisible to every per-KB query — and a later correct re-insert for the same (mention, entity) pair is **silently skipped by the unique key**: the orphan poisons the repair unless deleted first (production carries zero 0x00 rows today, so cleanup happened). The duration claim also fails row-level forensics: the registry's own `created_at` histogram (Figure 1) shows KB1 inserting at 1.0–2.3M rows/hour *through* the nominal silent window and KB2's entire link layer landing in one hour. We keep the log entry and its correction side by side: incident narratives written from job dashboards are themselves unverified claims.

7 EVALUATION

Entity layer. Concepts-per-document replaces the raw entity/document ratio: its numerator scales with documents and the coverage column exposes the collapse's most user-visible symptom — half the library had *no* annotations. Absolute link counts *rise* after repair (KB1 19.2M→25.1M; KB2 2.23M→5.0M): real terms match more often. Count was never the quality signal; distribution and names are.

Table 3: Entity layer before/after, with snapshot provenance.

Metric (KB1)	Collapsed	Repaired
canonical entities	575 / 503 (10-06)	**1,871**
entity links	~19.2M / 4.81M	**25.1M**
typed relations	~0	**44,043**
top-10 concentration	0.419	**0.345**
concepts/doc (all-doc)	15.7 (48.5% empty)	**37.8** (98.5%)
KB2: concepts/doc	29.5	**61.4**

Name quality (Figure 2). The repaired head is still generic: `be` (2.26M links, 9.0%), `alpha`, `gamma`, `beta`, `method`. The `be` case is a normalization failure — element symbol and auxiliary verb merge under case-folded matching — and §4 prices its treatment. The repair moved the problem from "no terminology at all" to "terminology plus a generic tail."

Link audit: 300 risk-stratified links. Entities drawn uniformly within four strata (top-10 by frequency; medium, ranks 11–100; tail, ranks 101+; short symbols, canonical names of ≤4 letters), then one link uniformly per entity — seeded and replayable. Span integrity 300/300. The judged-valid rate is **0.837** for this sample (Wilson 95% CI [0.791, 0.874]); by stratum: top-10 **0.920** [0.812, 0.968], medium 0.850 [0.767, 0.907], tail 0.910 [0.838, 0.952], short-symbol **0.580** [0.442, 0.706]. Three scope notes. First, the 50/100/100/50 quotas target risk, not production link frequencies: 0.837 is a stratum-mixed rate and must not be read as an unweighted production-wide link precision. Second, the 300 links come from 187 distinct entities (the head stratum's 50 links from just six; the short and head strata share "term"), so the intervals treat links, not concepts, as exchangeable. Third, the design's purpose is to localize high-risk failure — which it does: the short-ambiguous-symbol class that Figure 2 colors red and §4 prices is where validity collapses — a more informative finding than any overall figure. A second Qwen-series model re-adjudicated the earlier uniform sample: on the 55-link parseable subset agreement is 96.4% (domain 14/14) — a cross-model check, not family-independent confirmation (Figure 3), and unparseable responses may be non-random. Human adjudication remains the decisive future control.

Retrieval: honest attribution. Scoped coverage rose +6.6/+7.3 points over the repair window, but three controls bound the attribution: (a) removing the direct entity channels under the final configuration leaves @10 any-hit and document coverage unchanged on both scoped sets (MRR +0.0006, CI [0, 0.0018]; cov@20 −1.0pt on one set) while cutting p50 latency 1.46 s→0.52 s — and a forced-trigger check on the natural-phrasing set shows the channels are alive (54/99 fired; killing them costs 20 losses / 0 wins), so the ablation bounds these channels' marginal contribution in this configuration, not the semantic assets' value at large; (b) alias expansion, the one entity-adjacent mechanism, is +1.5pt (ns) on the benchmarks at 2× latency; (c) two same-configuration runs differ on exactly one query (0.5pt noise floor). The trajectory is system-level;

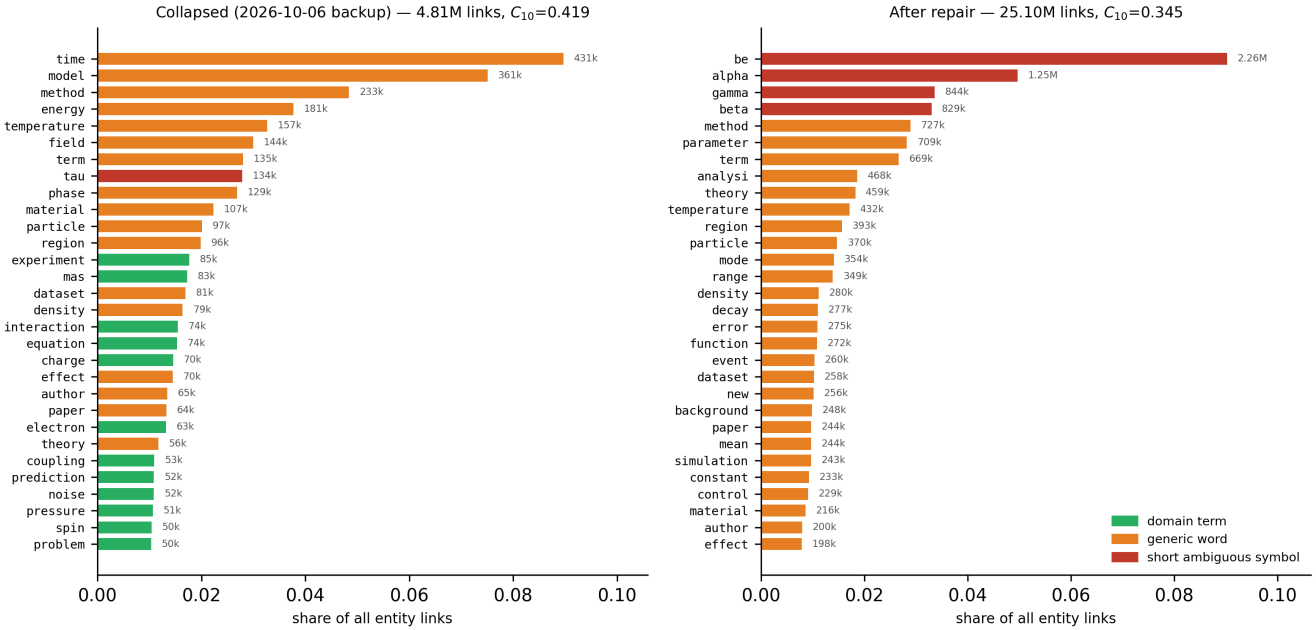


Figure 2: Top-30 entities by link share, before (cold backup; C_{10}=0.419) and after repair (C_{10}=0.345). Categories follow a fixed three-way rubric (domain term / generic academic word / short ambiguous symbol), assigned by the authors per entity; the rubric and per-entity assignments ship with the artifact.

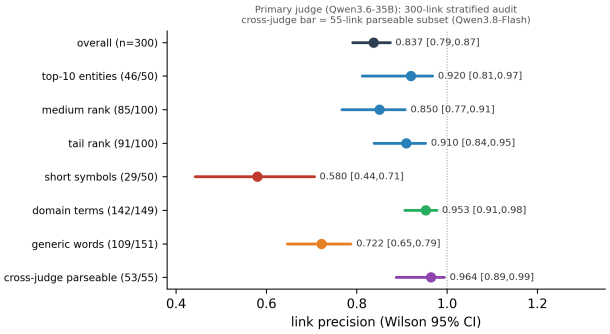


Figure 3: Judged-valid rates with Wilson 95% CIs: the risk-stratified 300-link sample (stratum-mixed, 187 entities; primary judge); the cross-judge bar applies to the 55-link parseable subset of the earlier uniform sample.

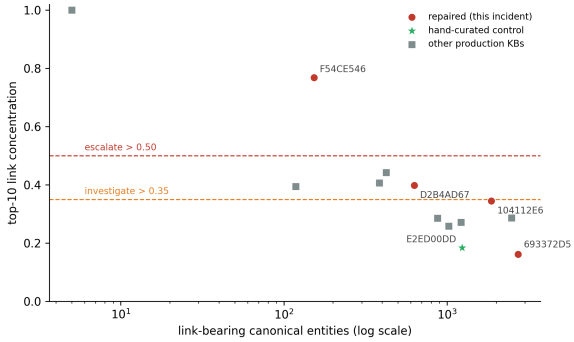


Figure 4: Top-10 link concentration vs. link-bearing entities across 13 production libraries; red = rebuilt in this incident, green star = hand-curated reference. Bands are candidate thresholds read post-hoc from this deployment.

the entity layer's retrieval-facing value runs through asset surfaces (documented in the companion system manuscript [7]), not fusion votes.

7.1 Monitors and Discussion

Four gold-free monitors, in the spirit of production data validation [4]: **(a) top-10 link concentration** (primary): invariant to uniform rescaling of counts, but bounded below by $\min(10, N)/N$ — trivially 1 for a library with ≤10 entities — and sensitive to topic concentration as well as to degradation. Across the 13 deployment libraries the metric spans 0.16–0.77 among libraries with ≥100 entities (the smallest, five entities, reads 1.000 — the floor bound, not degradation; Table 4); healthy/repaired large libraries run 0.16–0.19, the rebuilt nuclear library 0.345, the two small treated libraries *currently* 0.398 and 0.768 (reported as measured). The bands (>0.35 investigate, >0.5 escalate) are *candidate* thresholds read post-hoc from these 13 libraries — a starting point, not a validated norm. Validation need not re-collapse production: graded lexicon degradation injected offline into a frozen slice, reporting alarm precision, false-positive rate, detection delay,

Table 4: The candidate calibration behind Figure 4, summarized (the collapsed row is the pre-repair backup, not a live point).

Group	Libs	Entities	C_{10} range
Collapsed (backup)	1	503	0.419
Hand-curated control	1	1,239	0.184
Other prod., ≥500 ents	4	874–2,482	0.259–0.287
Rebuilt here, large	2	1,871 / 2,726	0.162–0.345
Rebuilt here, small	2	153–629	0.398–0.768
Other prod., <500 ents	4	5–424	0.395–1.000

and size sensitivity, is the natural protocol — future work here. **(b) entity cardinality ratio** (coarse secondary, gameable both ways). **(c) top-entity name audit** (five-minute quarterly control). **(d) target-table growth reconciliation** — the B95 ghost-bucket mechanism makes this mandatory wherever a bulk writer is guarded by `INSERT IGNORE`. The small-library treatment doubles as out-of-sample validation: cardinality normalized (870→629, 209→153) while concentration still flags both — the two monitors separate "cardinality repaired" from "distribution still skewed."

The incident is not an argument for LLM-everywhere: if extraction is a scan, the dictionary serves as the model and deserves a trained model's evaluation discipline. Admission criteria (corpus presence) are necessary but not sufficient; *ranking* criteria (IDF-style contrast, bilingual consistency, length/type priors) are the open problem — the generic tail is the standing evidence.

8 LIMITATIONS

(1) The generic tail survives repair (9% of links on "be"). (2) Retrieval attribution is bounded, not eliminated: the full lexicon×configuration decomposition is runnable on frozen slices or isolated stacks via graded degradation — not yet run, rather than impossible. (3) Coverage: two libraries rebuilt, two smaller treated; bilingual anomalies documented, not explained. (4) Six secondary incidents extended the repair; reported as findings. (5) Audit and cross-judge are model-assisted (same vendor family), not human-adjudicated; the risk-stratified sample mixes quota-defined strata and clusters on 187 entities, and unparseable cross-judge responses may be non-random. (6) The isolated experiment uses a semantic-replica scanner (not instrumented against the service path) and committed-snapshot dictionaries over ten resamples of one corpus — a stability check, not cross-domain evidence; its metrics are structural, with semantic reading confined to the sampled audit, and the governance arms await removed-vs-retained link verification.

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